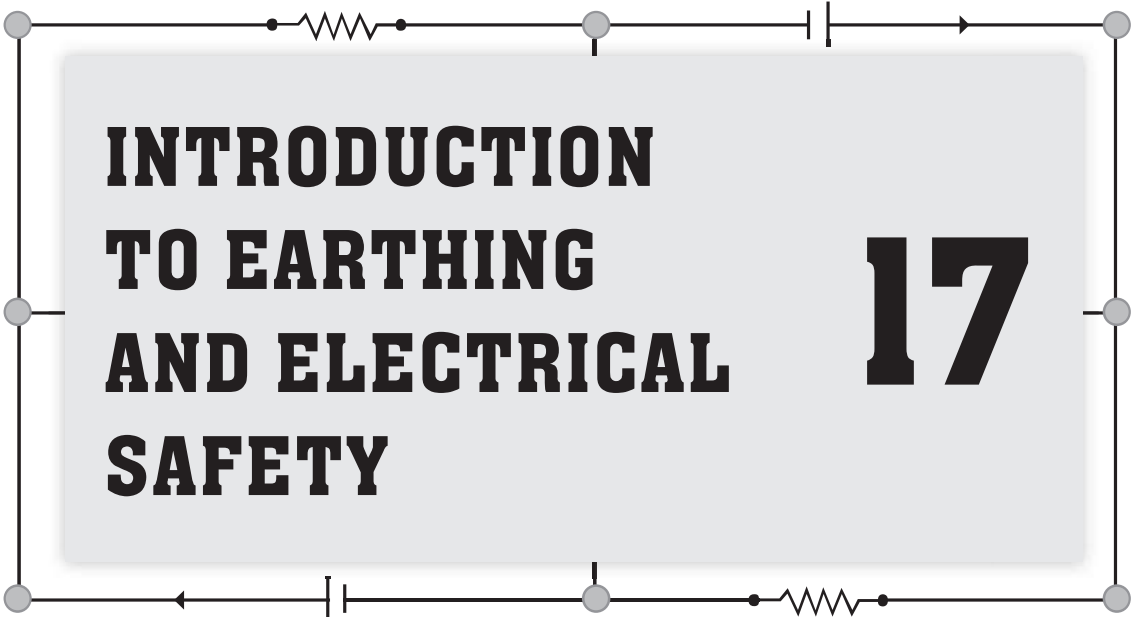


A decorative border around the title area, consisting of a rectangular frame with electrical symbols at the corners and midpoints. The symbols include resistors, voltage sources, and current arrows.

INTRODUCTION TO EARTHING AND ELECTRICAL SAFETY

17

LEARNING OBJECTIVES

After the completion of this chapter, the students or readers will be able to understand the following:

- ❖ What is an electric shock?
- ❖ When does a person experience an electric shock?
- ❖ How a victim of electric is treated?
- ❖ What precautions should be made against electric shock?
- ❖ What are the electric safety measures?
- ❖ What is earthing and its purpose?
- ❖ What are different methods of earthing?
- ❖ What is double earthing and its purpose?
- ❖ What is a fuse and an MCB? What are their roles in electrical installations?
- ❖ What is an ELCB or RCCB?

17.1 INTRODUCTION

The rapid industrial growth in twentieth century is mainly due to utilization of electrical energy in various fields. Now, electrical energy becomes a part of our day-to-day life.

Electricity is a good servant but a bad master if not handled properly. Therefore, while working with electrical installation and handling electrical equipment, one should always take care of his own and of other's safety. A little carelessness may result in an accident that may be fatal. Therefore, certain safety measures must be observed before dealing with electricity. In this chapter, various common safety measures and earthing of the equipments are discussed.

17.2 ELECTRIC SHOCK

When a person comes in contact with a live conductor, directly or indirectly, he gets a shock. The shock may be minor or severe. The severity of shock depends upon the following:

1. **Nature of the current whether AC or DC:** Since DC flows continuously and does not pass through zero current value as in AC, so DC is considered more dangerous than AC supply.
2. **Duration of flow of current:** Shock will be more severe if duration of contact with the live wire of a person is more. However, the severity of shock can be reduced by disengaging the person from live wire contact immediately.
3. **Path of current through human body:** Severity of shock also depends upon the path of the electric current through human body. A person has severe shock if current path involves his heart.

The effect of electric current when passes through human body is given in Table 17.1.

Table 17.1 Effect of Electric Current Through Human Body

1–10 mA	Prickling sensations
10 mA	Muscle contraction: The person remains 'stuck' to the conductor
20–30 mA	Muscles contraction can cause respiratory paralysis
70–100 mA	Cardiac fibrillation: The heart begins to vibrate and no longer beats at steady rate. This situation is dangerous since it is irreversible.
500 mA	Immediate cardiac arrest resulting in death

However, electrocution should not be viewed in terms of 'current' alone but in terms of 'current and voltage'. A person gets electrocuted by coming in contact with an object that has a different potential from his/her own. The difference in potential causes the current to flow through the body.

The human body has known limits:

Under normal dry conditions, voltage limit = 50 V

In damp surroundings, voltage limit = 25 V

17.3 ELECTRIC SHOCK TREATMENT

The victim of electric shock should be immediately treated as suggested below:

1. Victim should be removed immediately from the contact of live conductor.
2. Artificial respiration should be given immediately.
3. Treat the burns, if any, on recovery of the victim.
4. Finally, give call to a doctor.

For removing the victim from the contact of L.T. (Low Tension) live wire, any one of the following procedures can be followed:

1. Immediately 'Switch OFF' the supply. If switch is far away, then pull out the plug top.
2. Pull the victim by using wooden stick, dry clothes, dry rope, etc.
3. One can pull victim directly by standing on well-insulated footing such as rubber mat, dry board, dry wooden chair, electrician rubber gloves or even pull directly (to some extent) if wearing rubber sole shoes.
4. Cut the conductor by an axe, or axe like device having a large wooden handle.

For removing the victim from the contact of HT (high tension) live conductor, any one of the following procedures can be followed:

1. 'Switch OFF' the circuit breaker, if it is nearby.
2. Short the live conductors by throwing a bare wire or chain upon them. This will result in tripping of the circuit breaker, at the substation or power station, as the case may be.

17.4 METHODS OF ARTIFICIAL RESPIRATION

Once the victim has been removed from the contact of the live wire, next step is to give him artificial respiration. There are various methods of giving artificial respiration, which are detailed in this section.

17.4.1 Schafer's Method

The Schafer's method is the best method to give artificial respiration to the victim of electric shock. The various steps of this method are as follows:

1. Victim is laid on the stomach, with his face on one side and pull the arms forward as shown in Figure 17.1(a).
2. To allow proper breathing, victim's neck is cleared from clothing.
3. Clear the mouth of the victim from pan, tobacco, artificial teeth, etc.
4. Kneel over the victim as shown in Figure 17.1(b) and place both your hands flat on his back. Place your hand in such a manner that both of your thumbs nearly touch each other and fingers are spread on each side of the victim's lower ribs.

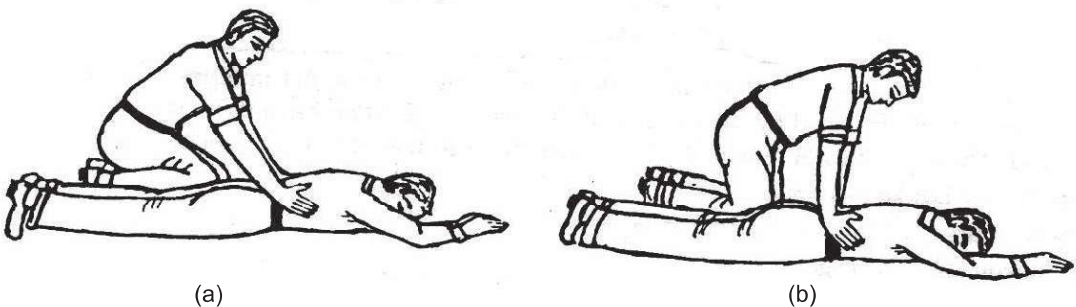


Fig. 17.1 Schafer's method of artificial respiration (a) Lay the victim on the stomach (b) Kneeling over the victim

5. Lean forward over the victim gradually and gently by putting your weight on the victim for a moment as shown in Figure 17.1(b).
6. Now slowly release the pressure and come to original position.

Repeat the process at least 12 to 15 times in one minute, till the victim starts natural breathing.

Schafer's method of artificial respiration is better method as compared with other two methods to be followed. Hence, this method should be adopted as and when required. However, this method cannot be used if it is not possible due to burns to lay the victim on his stomach.

17.4.2 Silvestre's Method

Schafer's method is the best method of artificial respiration; however, if the victim got some burns on the chest or anywhere on the front side of the body, only then this method is adopted. This method can be proceeded as follows:

1. Lay the victim on his back, as shown in Figure 17.2(a).
2. Remove the victim's clothes around his neck.
3. Clear the mouth of the victim from pan, tobacco, artificial teeth, etc.
4. Use pillow or rolled coat or any other cloth under the shoulders of the victim, so that his head falls backwards.
5. Tilt the head a little back. It will keep the tongue out of the throat allowing passage to air for breathing.
6. Now kneel in the position near the head of the victim as shown in Figure 17.2(b).
7. Stretch both arms of the victim backward by holding them below the elbows.
8. Keep the arms in this position for about 2 to 3 s.
9. Now bring the arms of the victim on each side of his chest (Fig. 17.2(b)), so as to compress his chest. Keep the victim's arms in same position also for 2 to 3 seconds.
10. Repeat the process for about 12 to 15 times in a minute, till victim starts natural breathing.

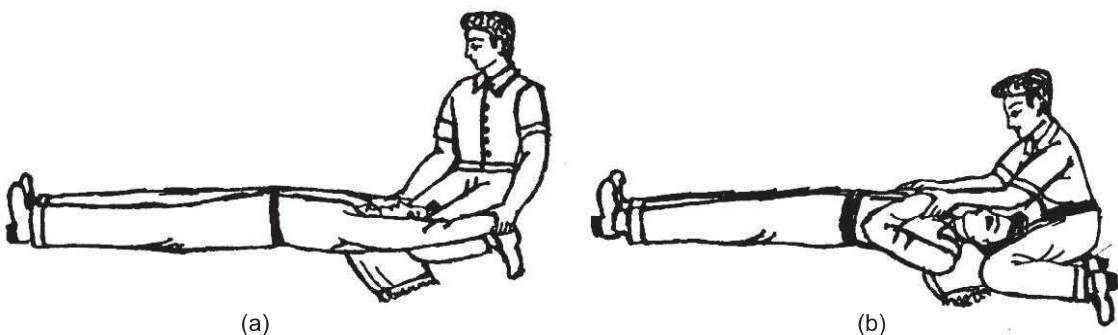


Fig. 17.2 Silvestre's method of artificial respiration (a) Lay the victim on the back (b) Kneeling near the victim's head

It should be assured that this method is to be adopted only when it is not convenient to make the victim lie on his stomach due to burns.

17.4.3 Third Method (Artificial Respirator Method)

It is the easiest and most hygienic method of artificial respiration, if the apparatus is available. When the victim has suffered an electric shock and is unconscious, an artificial respirator may be used. An artificial respirator consists of a rubber bulb mask and an air filter along with a transparent celluloid valve arrangement as shown in Figure 17.3(b). The air enters through the holes of rubber bulb and goes out through the outlet valve. The mask is placed on the mouth and nose of the patient as shown in Figure 17.3(a). The rubber bulb is pressed at the rate of 12 to 15 times per minute to bring his respiration back. This process should be continued regularly till the doctor advises to stop.

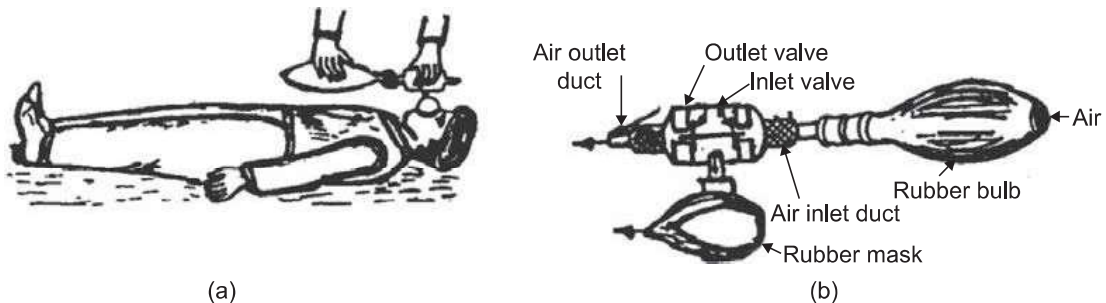


Fig. 17.3 (a) Artificial respiration (b) Respirator

The working principle of respirator is as follows: The rubber mask is fitted on the mouth and nose of the victim. When the rubber bulb is pressed, the air of the bulb passes through the air filter that lifts the inlet valve and closes the outlet valve. Now, this filtered air enters the lungs of the patient through the mask and nose. When the pressure on the bulb is released, the inlet valve closes and the outlet valve is opened, which gives path to the used air to go out.

Special Instructions for treatment are as follows:

1. Never give any drink to the unconscious man.
2. Violent operation of the process must be avoided because an internal injury in the affected organ may be harmed due to quick and excessive pressure.
3. If there is a burn on the body, it should be properly dressed after the recovery of the patient.
4. The patient should be kept warm.
5. No medicine should be given without the consent of the doctor.
6. An owner of the factory must provide and fix a chart explaining the methods of artificial respiration. The chart should carry the name of the nearest doctor and his telephone number. Preferably, the address of the hospital and residential address of the doctor to be contacted should be given in the chart.

17.5 PRECAUTIONS AGAINST ELECTRIC SHOCK

The following precautions should be taken as preventive measures from electric shock while dealing with electrical equipment fittings or appliances:

1. Never work on live circuit.
2. Always stand on the insulating material, such as rubber mat, wooden board, etc., while switching on the main switch, motor switch, etc.
3. While switching ON the circuit, equipment, etc., ensure that your hands and feet are dry.
4. Avoid working at all those places where your head is liable to touch the live parts.
5. While working with electrical circuits/equipments, never come in contact with the metallic casing, earth conductor, cross arms, etc.
6. While working on the high-voltage circuit, avoid your direct contact with concrete flooring.
7. Never touch the person directly, while rescuing him from electric shock.
8. Consider all conductors as live, till you are not sure.

17.6 ELECTRIC SAFETY MEASURES

The following common safety measures should be followed while dealing with the electricity:

1. Always follow IER (Indian Electricity Rules) while dealing with electrical equipment and installations.
2. Consider all conductors (insulated or bare) as live conductors. So do not touch them till you are not sure.
3. Switch OFF the main switch and keep the fuse carriers with you while working on an electrical installation.
4. Single-way switches should be always connected in a live wire.
5. Fuse should be only provided in a live wire.
6. Before replacing the blown fuse or switching ON the tripped MCB (miniature circuit breaker), be sure that the defect has been rectified.
7. Do not work on live circuits. However, wherever the energized circuits are to be handled, then always use rubber mats, rubber sole shoes, rubber gloves, etc., as the case may be.
8. Never disconnect the appliance from the plug point by pulling the connecting cord.
9. Always use proper size of wire of proper voltage grade for all electrical appliances, equipments, wiring installation, etc.
10. Always use standard cable for connecting portable equipments, appliances, pendant holders, etc. Since standard conductor cable provides flexibility, so equipment can be handled conveniently.
11. All electrical connections should be made tight and these should be checked periodically, so as to avoid fire due to loose connections.
12. Always use water proof cables and fittings for all outdoor installations except distribution and transmission lines.
13. All non-current carrying metal parts of the equipment and of installation should be properly earthed.

14. All portable electrical equipments should be properly earthed. Therefore, for such equipments, always use three-core cable.
15. Always keep the earthing in good condition, that is, earth resistance should be kept very low all the time, since safety depends upon perfect earthing.
16. As far as possible, put off the main switch (or controlling switch), when a person is still in contact with a live conductor.
17. Do not disengage a person, who is still in contact of live circuit, by touching him directly. Push him only with a piece of dry wood or other such insulating material.
18. In case, a person is electrocuted, immediately apply artificial respiration and call the doctor.
19. Never use water for extinguishing fire due to electric current. Use only carbon tetrachloride, liquid carbon dioxide fire extinguishers. Sand can also be used for extinguishing the electric fire.

17.7 EARTHING

The process of connecting metallic bodies of all the electrical apparatus and equipment to the huge mass of earth by a wire of negligible resistance is called earthing.

When a body is earthed, it is basically connected to the huge mass of earth by a wire having negligible resistance. Therefore, the body attains zero potential, that is, potential of earth. This ensures that whenever a live conductor comes in contact with the outer body, the charge is released to the earth immediately.

17.7.1 Purpose of Earthing

The basic purpose of earthing is to protect the human body (operator) from electric shock. To illustrate the purpose of earthing, consider an electrical circuit shown in Figure 17.4 where an electrical appliance of resistance R is connected to the supply through a fuse and a switch. When an operator touches the metallic body of the apparatus [Fig. 17.4(a)] having perfect insulation, the equivalent circuit is shown in Figure 17.4(b), where two parallel paths are formed. Since the insulation resistance R_i is very high as compared with appliance resistance R , whole current flows through appliance resistance and no current flows through human body (operator's body) resistance.

When earth fault occurs, the live (phase) wire directly comes in contact with the outer body and the insulation resistance reduces to zero as shown in Figure 17.4(c). Now, the body resistance is just in parallel with the appliance resistance. A heavy current flows through the human body and operator gets a severe shock.

However, if the metallic body or outer frame of the appliance is properly earthed, then under earth fault condition, the circuit will be as shown in Figure 17.4(d), where earth resistance R_e is just in parallel with the appliance resistance R and body resistance R_b . Since earth resistance is very small as compared with body resistance, almost whole of the fault current flows through the earth resistance and no current flows through the human body. Therefore, the operator is protected from electric shock. Moreover, the fault current is much more than the full-load current of the circuit that melts the fuse. Hence, the appliance is disconnected automatically from the supply mains.

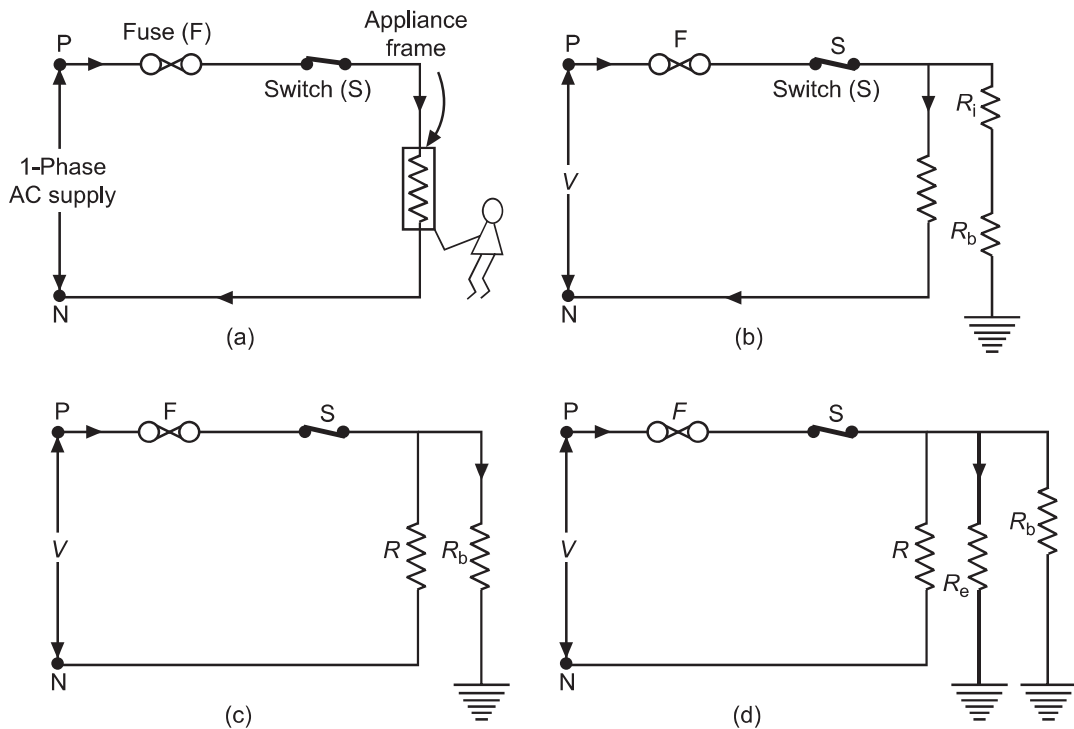


Fig. 17.4 (a) Operator touching the metallic body of the apparatus (b) Under normal condition insulation and body resistance come in series (c) When frame comes in contact with live wire, the insulation resistance vanishes (d) When earthing is provided, low earth resistance come in parallel with body resistance

17.7.2 Equipment Earthing

According to Rule 61 of Indian Electricity Rules 1956, it is obligatory to earth by the following points and apparatus used in the power system, where the voltage is more than 125 V:

1. All the metal frames of motors, generators, transformers, and controlling equipment.
2. The steel tower and steel tubular or rail poles carrying overhead conductors.
3. The metal frames of portable electrical equipment such as heaters, table fans, electric iron, refrigerator, air conditioners, vacuum cleaners, etc.
4. Other metal parts such as conduits, switch gear casings, etc.
5. Earth terminal of all the three-pin outlet sockets.
6. In case of concentric cables, external conductor, that is, armouring of such cables.
7. Stay wires of overhead lines if stay insulator is not provided.

In case of insulation failure, the primary objective of connecting all the above points and apparatus to earth is to release the charge accumulated on them immediately to earth so that the person coming in contact may not experience electric shock. The other objective is that a heavy current when flows through the circuit operates the protective device (i.e., fuse or miniature circuit breaker), which opens the circuit.

Generally, the values of earth resistance given in Table 17.2 must be achieved while earthing.

Table 17.2 Value of Earth Resistance for Different Equipment

Equipment to be earthed	Maximum Value of Resistance Under Worst Conditions
1. Large power stations	0.5 Ω
2. Major substations	1.0 Ω
3. Small substations	2.0 Ω
4. Factories substations	1.0 Ω
5. Lattice steel towers	3.0 Ω
6. Industrial machines and equipment	0.5 Ω

The earth resistance depends upon the moisture contents in the soil and varies from month to month. Therefore, earth resistance must be checked periodically by earth tester and maximum permissible value be obtained by pouring water into the funnel.

17.7.3 System Earthing

A proper system has to be adopted while earthing. In fact, all the heavy power equipment should be earthed by two separate distinct earth wires following the different routes. The two earth connections are applied to improve the reliability. If one of the earth wires breaks or fails to carry the fault current, the other carries that current and provides the required protection. Moreover, in factories and substations, where more than one equipment is to be earthed, parallel connections should invariably be used. In no case series, connections are done, as even a single bad contact or break in the earthing lead will disconnect all the succeeding equipment from the earth.

Therefore, for proper earthing of heavy power equipment, double earthing system has to be adopted. Moreover, the number of apparatus must be connected in parallel to the earth.

17.7.4 Methods of Earthing

As discussed earlier, earthing means to connect metallic bodies of the apparatus with the general mass of earth by a wire of negligible resistance. There are various methods of achieving this connection, some of them are given below:

1. **Strip earthing:** This system of earthing employs the use of 5 SWG (standard wire gauge) copper wire or strip of cross section not less than 25 mm $\times$ 1.6 mm. The strips or wires are buried in horizontal trenches. This type of earthing is used where the earth bed has a rocky soil and excavation work is difficult.
2. **Earthing through water mains:** In this type of earthing, a stranded copper lead is used that is rounded on the pipe with the help of a steel binding wire and a properly designed earthing clip. The copper lead is soldered to make it solid. Before making connection to the water main, it should be ensured that G.I. pipe is used throughout.
3. **Rod earthing:** It is the cheapest method of earthing and is employed in sandy areas. In this method, a copper rod is hammered directly into the ground, and no excavation work is required. The earthing lead is joined to this rod with the help of nuts and bolts.
4. **Pipe earthing:** Taking into consideration, the factors such as initial cost, inspection, resistance measurement, etc., G.I. pipe earthing is best form of ground connection. Iron is the cheapest material and remains serviceable even if put in salty mass of earth. The

pipe used as earth electrode is galvanized and perforated. Its diameter is 38.1 mm and length is 2 m. The length may be increased to 2.75 m in case of dry soil. The diameter of pipe has very little effect on the resistance of the earth connection. To facilitate the driving in of the pipe into ground, it is provided with the tapered casting at the lower end. Another pipe of 19.05 mm diameter and of length 2.45 m is connected at the top of the above-mentioned pipe. The connection between these pipes is made through a reducing socket as shown in Figure 17.5.

The earthing lead should be soldered and connected to the pipe. Alternate layers of charcoal and salt are provided around the G.I. pipe to keep the surroundings moist enough. The salt is poured at the bottom and thereafter alternate layers of charcoal and salt are arranged.

5. **Plate earthing:** In this type of earthing, a copper or G.I. plate of dimensions not less than $60 \text{ cm} \times 60 \text{ cm} \times 3.18 \text{ mm}$ or $60 \text{ cm} \times 60 \text{ cm} \times 6.35 \text{ mm}$ is used as earth electrode instead of G.I. pipe. The plate is buried into ground in such a way that its face is vertical

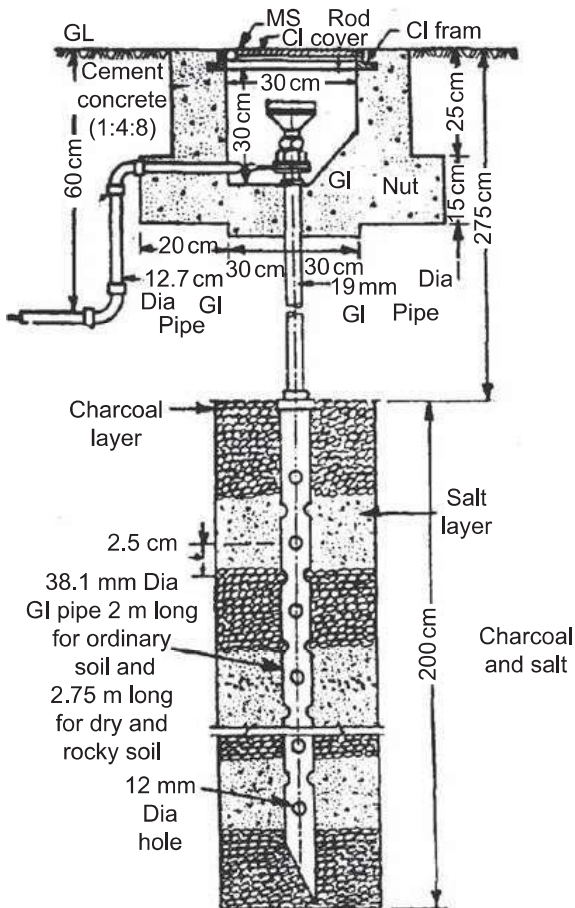


Fig. 17.5 Pipe earthing

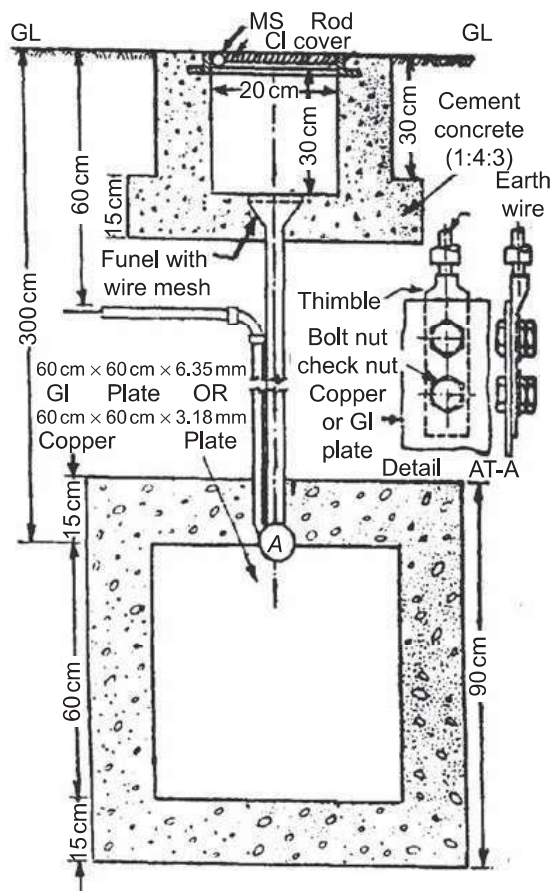


Fig. 17.6 Plate earthing

and the top is not less than 3 m below the ground level. The G.I. wire is used for G.I. plate and copper wire for copper plate earthing. The size of wire is selected according to the installation and fault current. The earthing lead is suitably protected placing it underground in a pipe as shown in Figure 17.6.

Alternate layers of charcoal and salt are used around the plate. The layers of charcoal shall be placed immediately over the plate, and thereafter, successive layers of salt and charcoal are laid to keep the surroundings sufficiently moist.

Note: Pipe earthing and plate earthing are considered to be the best as they have reasonably low value of earth resistance.

17.8 SIZE OF EARTH WIRE

In case of household wiring or installation, a 14 SWG hard drawn bare copper conductor is used as earth wire. For power installations, the size of earth wire depends upon the rating of the motors installed. For different ratings of motors, the size of earth wire can be selected from Table 17.3.

Table 17.3 The Size of Earth Wire for the Motors of Different Rating

Capacity of Apparatus	Size of Earth Wire in SWG		Size of Earth Electrode	
	Copper	G.I	Copper	G.I
Up to 10 H.P.	No. 8	No. 8	60 cm × 60 cm × 3.18 mm	60 cm × 60 cm × 6.35 mm
Above 10 H.P. and up to 15 H.P.	No. 8	No. 6	—do—	—do—
Above 15 H.P. and up to 30 H.P.	No. 6	No. 2	—do—	90 cm × 90 cm × 6.35 mm
Above 30 H.P. and up to 50 H.P.	No. 4	—	90 cm × 90 cm × 6.35 mm	—do—
Above 50 H.P. and up to 100 H.P.	No. 2 or strip 12.7 × 2.5 × 4 mm	—	—do—	—do—
Above 100 H.P.	Strip 25.4 × 2.5 × 4 mm	—	—do—	—do—

The conductor used for earthing purpose should not be of cross section less than 14 SWG and greater than 64.52 sq. mm from mechanical considerations. From electrical considerations, the copper earth wire should not be of size less than half of the largest current carrying conductor.

17.9 DOUBLE EARTHING

For providing better safety, it is advisable to provide two separate earth wires, from two separate earth electrodes, connected to same metallic body of the equipment at two different points. This is known as double earthing. Double earthing is essential, as per Indian Electricity Rule, for metallic bodies of large rating equipment such as transformer, motors, etc. working at 400 V and above.

Advantages of double earthing are as follows:

1. Surety of safety, because if at any time, one earthing is ineffective, then another will provide earth path to fault current.
2. As the two earth wires are in parallel, the effective resistance from equipment to earth electrode is reduced.



PRACTICE EXERCISES

Short Answer Questions

1. What do you mean by electric shock?
2. How to proceed with the electric shock treatment?
3. What are the rules that suggest the safety measures?
4. What is earthing?
5. What is the purpose of earthing?
6. What do you understand by equipment earthing?
7. What do you understand by system earthing?
8. What are the different methods of earthing?
9. What do you mean by double earthing?

Test Questions

1. Under what conditions, a person experiences an electric shock?
2. For electric shock treatment, what are the methods of artificial respiration? Explain any one in detail.
3. Explain Silvester's method of artificial respiration.
4. Mention 10 electric safety measures.
5. What is earthing? How does it protect a person from electric shock?
6. How will you differentiate between equipment earthing and system earthing?
7. What are the different methods of earthing? Explain plate earthing with a neat sketch?
8. Name different methods of earthing and explain pipe earthing with a neat sketch.

17.10 CAUSES OF ELECTRIC FIRE

The following are the causes of electric fire:

1. Use of inferior quality of materials.
2. Overloading of the circuits.
3. Proper protective devices are not employed.
4. Insulation of the wires is damaged.
5. When bare conductor come in contact with each other.
6. When bare conductors come in contact with the earthed points in the electrical installation.
7. Loose connections.

17.11 PREVENTION OF ELECTRIC FIRE

Fire due to electric current can be prevented by taking the following precautions:

1. Use superior quality of materials in the electrical installation and with each and every equipment.
2. By using protective devices of proper rating with the electrical circuits, so that when excessive current flows due to sustained overloading, short circuit fault or earth fault, the

protective devices should operate for disconnecting the supply to faulty circuit/equipment. This prevents damage to installation/equipment and danger of fire.

3. Overloading of every equipment and electrical installation should be avoided.
4. Well insulated and proper size of wires should be used.
5. The joints in the electrical system should mechanically and electrically be sound. For electrical installation, always use looping in system instead of jointing in system of wiring. There should not be any loose connection in the installation.
6. All connections in the electrical installation should be tight and these should be checked periodically, especially when aluminium wires have been used in the installation.
7. The electrical installation should be free from moisture effect. Wiring route should not be near to the water pipe installation.
8. In case of electric fire, never use water when circuit is live. To avoid the spreading of electric fire and for rescue operation, following fire fighting equipment should be kept ready for use.
 - (a) Fire extinguisher carbon tetra chloride (i.e., CTC) or foam type
 - (b) Fire extinguisher carbon dioxide

These fire extinguishers employ chemicals that are insulators and do not give shock when thrown on to the live parts. Hence, these are used for extinguishing fire due to electrical short circuits and sparks.

1. Buckets filled with sand
2. Ladder
3. Rope
4. Stretcher
5. First aid box, etc.

17.12 FUSE

A short piece of metal wire, inserted in series with the circuit, which melts when predetermined value of current flows through it and breaks the circuit is called a fuse.

A fuse is connected in series (Fig. 17.7) with the circuit to be protected and carries the load current without overheating itself under normal conditions. However, when abnormal condition occurs, an excessive current (more or equal to the predetermined value for which the fuse is designed) flows through it. This raises the temperature of the fuse wire to the extent that it melts and opens the circuit. This protects the machines or apparatus from damage that can be caused by the excessive current.

Time–current characteristics: The time required to blow out a fuse depends upon the magnitude of excessive current. Larger the current, smaller is the time taken by the fuse to blow out. Hence, a fuse has inverse time current characteristic as shown in Figure 17.8, which is desirable for a protective device.

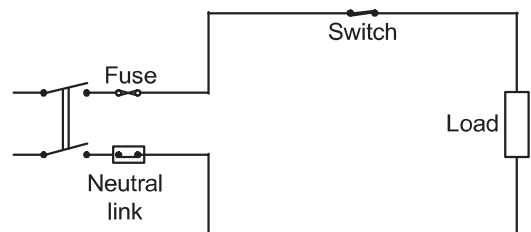


Fig. 17.7 Electric circuit with fuse and switch

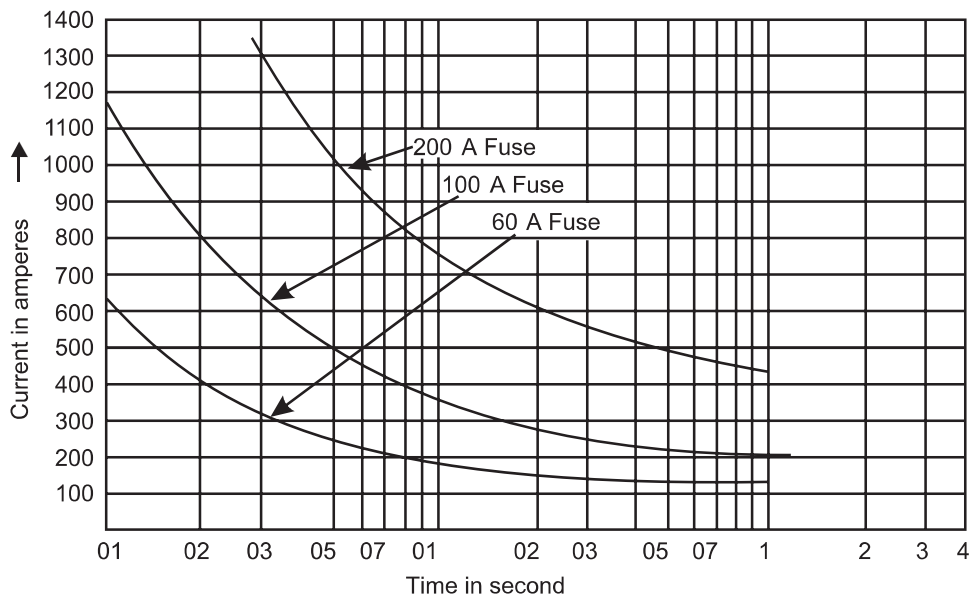


Fig. 17.8 Typical characteristics of 60A, 100A and 200A fuse

17.12.1 Advantages of Fuse

1. The cost of this protective device is very low.
2. It requires no maintenance.
3. It interrupts heavy current without noise or smoke.
4. The smaller size of fuse element imposes a current-limiting effect under short circuit.
5. The minimum time of operation can be predetermined by selecting proper material of the fuse wire.
6. The inverse time–current characteristic makes it suitable for over current protection.

17.12.2 Disadvantages of Fuse

1. Considerable time is lost in rewiring or replacing fuses after every operation.
2. On short circuit, determination between fuses in series can only be obtained if there is considerable difference in the relative sizes of the fuse concerned.

17.13 MINIATURE CIRCUIT BREAKER (MCB)

Miniature circuit breaker (MCB) is a device that ensures definite protection of wiring system and sophisticated equipment against over current and short circuit faults. The outer view and the internal details of a MCB are shown in Figures 17.9 and 17.10, respectively.



Fig. 17.9 Outer body of a miniature circuit breaker

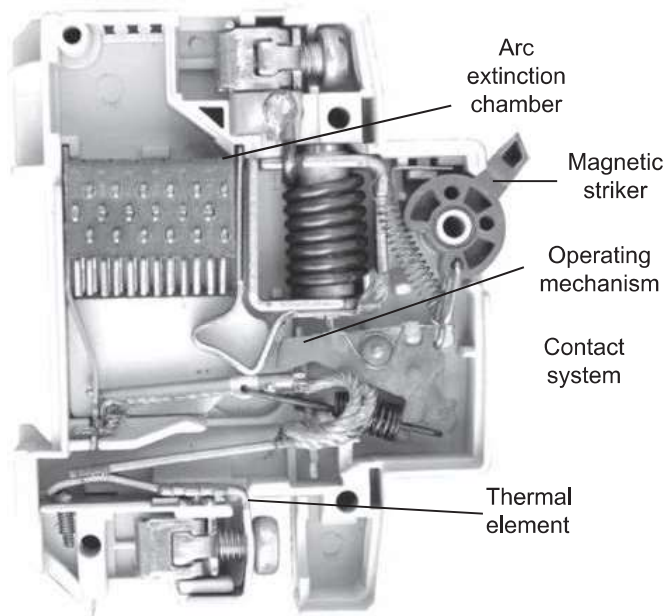


Fig. 17.10 Internal parts of a miniature circuit breaker

17.13.1 Construction

Construction of an MCB can be explained by considering the following main parts:

1. **Outer body or housing:** The outer body or housing of an MCB is moulded from a special grade glass fibre-reinforced polyester with the help of an injection-moulding machine. The outer body and other polyester components of MCB are fire-retardant, anti-tracking and non-hygroscopic. These polyester parts and housing have the ability to withstand high-temperature and mechanical impacts.
2. **Contacts:** The contacts of an MCB are made of pure silver. This provides definite advantages such as long contact life, low contact resistance, ensures quick arc removal, and low-heat generation.
3. **Operating mechanism:** All the components of the operating mechanism are made of special plastic that they are self-lubricating that eliminates wear and tear, rust, and corrosion. These components are very light in weight and have low inertia, thereby ensure snap make the break ability. The reliability and ruggedness of the operating mechanism is, thus, maintained.
4. **Arc extinguishing chamber:** The arc produced during breaking of circuit is extinguished abruptly by providing a special arc chute chamber.
5. **Fixing arrangement:** The MCB-mounting clip gets easily snapped on to the Din-bar and can be removed easily by a simple operation with a screw driver. This saves the time that would have been required for fixing it with screws.

6. **Mechanical interlocking of multiple MCBs:** The levers of all the (3 or 4) multiple MCBs are connected internally. This ensures simultaneous tripping of all poles even if the fault develops in any one of the phases.

17.13.2 Working

MCB may operate under the following two different conditions:

1. **Moderate overload condition:** Detection of moderate overload conditions is achieved by the use of a thermometal that deflects in response to the current passing through it. The thermometal moves against the trip lever, releasing the trip mechanism.
2. **Short circuit conditions:** When the current flowing through the MCB reaches a predetermined level (as per its setting or rating), it pushes the solenoid plunger that releases the trip mechanism and simultaneously separates the contacts.

Under short circuit conditions, the current-limiting action is achieved by the use of a high speed, direct acting electromagnetic mechanism.

This mechanism forcibly separates the contacts and simultaneously releases the trip mechanism. A high arc voltage drop is rapidly introduced that limits the fault current to a duration of few milliseconds and achieves almost instantaneous interruption (the facts are shown graphically in Fig. 17.11).

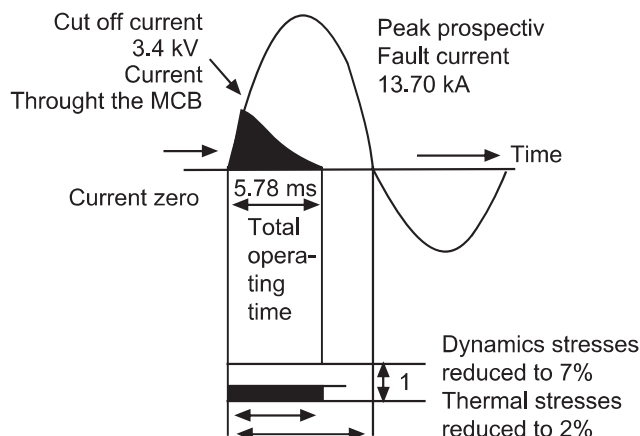


Fig. 17.11 Operating time of an MCB

When the contacts are separated, the current still rises due to arc. This arc is extinguished quickly in the arc chute chamber and does not allow the current to reach theoretical maximum value. The total breaking time is reduced to less than 5 millisecond.

17.13.3 Applications

Since MCBs are available with different current ratings of 0.5, 1.6, 2, 2.5, 3, 4, 5, 6, 7.5, 10, 16, 20, 25, 32, 35, 40, and 63 A and voltage ratings of 240/415 V AC and up to 220 V DC. Moreover, they have very small breaking time (5 millisecond), and therefore, these are generally employed to protect the important and sophisticated appliances used commercially and

for domestic purposes, such as computers, air conditioners, compressors, refrigerators, and many others.

17.14 EARTH LEAKAGE CIRCUIT BREAKER (ELCB)

In the industrial, commercial, and domestic buildings, sometimes (usually in rainy season) leakage to earth occurs. This leakage may cause electric shock or fire. Hence, the leakage to earth is very dangerous and needs protection.

ELCB is a device that provides protection against earth leakage faults.

17.14.1 Construction and Internal Circuit Details

The enclosures of the ELCB is moulded from high-quality insulating material. The materials are fire-retardant, anti-tracking, non-hygroscopic, impact resistant and can withstand high temperatures. The body contains spring-loaded mounting arrangement on din-channel that ensures snap fitting of ELCB into position. However, these also have the facility to screw-on directly to any surface with the help of two screws. A two-pole ELCB is used for single-phase supply and a four-pole ELCB is used for three-phase, four-wire supply. A four-pole ELCB is shown in Figure 17.12.

Internal wiring diagram of a two-pole ELCB is shown in Figure 17.13. As shown in Figure 17.13, an ELCB contains a core-balanced transformer (ferrite ring on which one or two turns of phase and neutral wire, and a few turns of operating coil of relay are wound) and a relay. A test button is placed between phase and neutral in series with a limiting resistor. The terminal designation and connection diagram for a two-pole and four-pole ELCB are shown in Figures 17.14 and 17.15, respectively.



Fig. 17.12 Outer body of an earth leakage circuit breaker

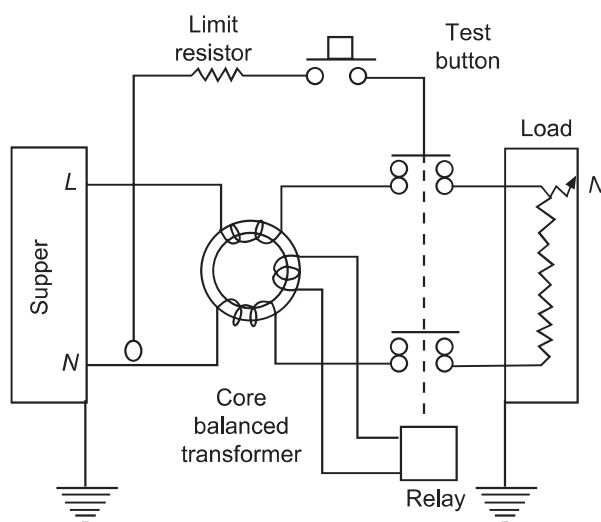


Fig. 17.13 Electrical circuit of an earth leakage circuit breaker

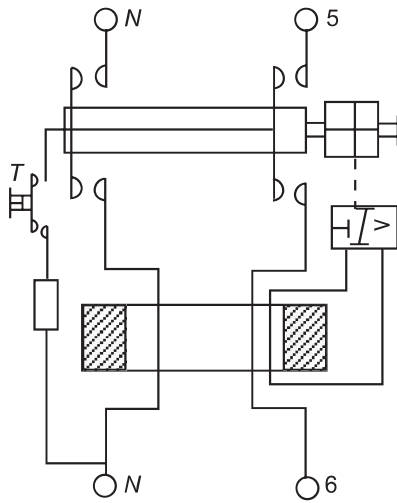


Fig. 17.14 Schematic diagram of a 2-pole ELCB

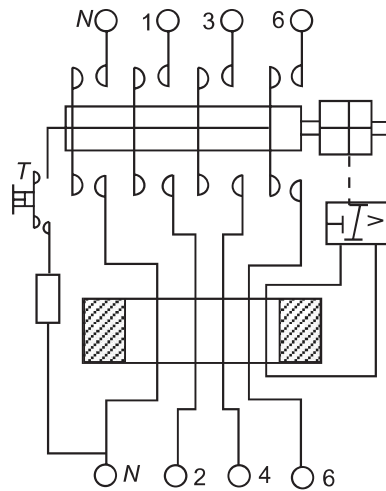


Fig. 17.15 Schematic diagram of a 4-pole ELCB

17.14.2 Principle of Operation

Under normal conditions, the magnitude of currents flowing through the phase wire and neutral are the same and core of the core-balanced transformer does not carry any flux (i.e., two windings neutralize the flux). Therefore, no emf is induced in the operating coil of the relay wound on the same core. However, when the earth fault (earth leakage) occurs, the current in the phase wire becomes more than the neutral wire. This unbalancing sets up flux in the core of the core-balanced transformer, which in turn induces an emf in the operating coil of the relay. Hence, relay is energized and the plunger of the ELCB goes to the off position or disconnects the load from the supply. Therefore, ELCB protects the system against leakage.

Use of test knob

A test knob is provided for the periodic checking of the mechanism and function of ELCB.



PRACTICE EXERCISES

Short Answer Questions

1. What are the causes of electric fire?
2. What is a fuse?
3. What are different types of fuses?
4. What is MCB? How does it work?
5. What is the full form of RCCB and its usage in wiring system?

Test Questions

1. How electric fire can be prevented?
2. What is an electric fuse? Draw the characteristics of a fuse and mention its merits and demerits.
3. What are the different parts of an MCB and explain how it works.
4. Explain the working of an ELCB.

SUMMARY

1. *Electric shock*: When more than 10 mA current passes through a human body, one experiences an electric shock.
2. *Electric shock treatment*: When a person experiences an electric shock, first of all he/she should be removed from the electric contact and then artificial respiration should be given to the patient till doctor reaches at the spot.
3. *Safety measures*: Indian electricity rules have suggested various safety measures while dealing with electricity.
4. *Earthing*: The process of connecting metallic bodies of all the electrical apparatus and equipment to the huge mass of earth by a wire of negligible resistance is called earthing.
5. *Purpose of earthing*: The basic purpose of earthing is to protect the human body (operator) from electric shock.
6. *Equipment earthing*: According to Indian electricity rules, it is obligatory to earth the metallic bodies of all the electrical equipment/apparatus/appliance that is operated at 125 V or more than this.
7. *System earthing*: All the electrical equipment/apparatus/appliance operated at 125 V or more must be properly earthed at least at two places called double earthing.
8. *Methods of earthing*: The following method may be employed for earthing (i) strip earthing (ii) earthing through water mains (iii) rod earthing (iv) pipe earthing (v) plate earthing. Pipe earthing is considered the best method.
9. *Causes of electric fire*: Electric fire may be caused by (i) using inferior quality material and (ii) due to overloading of circuit. (iii) Use of improper protective devices. (iv) Poor quality of insulation of wiring cables. (v) Loose connection, that is, sparking. (vi) Bare conductor coming in contact with earth.
10. *Fuse*: A short piece of metal wire, inserted in series with the circuit, which melts when more than predetermined value of current flows through it and breaks the circuit is called a fuse.
11. *Types of fuses*: (i) Low-voltage fuses, that is, (a) rewirable fuses (b) high breaking capacity fuses (HBC), or high-rupturing capacity (HRC) fuses (iii) high-voltage fuses, that is HBC fuses.
12. *Miniature Circuit Breakers (MCBs)*: It is a protective device that is connected in series with load in phase wire. It trips or disconnects the load as and when current through it flows more than its predetermined value.
13. *Earth Leakage Circuit Breaker (ELCB) or Residual Current Circuit Breaker (RCCB)*: It is a protective device that disconnects the circuit or load from the mains when earth fault develops in the circuit.

TEST YOUR PREPARATION

FILL IN THE BLANKS

1. One experiences an electric shock when the current passing through one's body is more than _____.
2. A short piece of wire connected in series with the circuit that melts when excessive current flows through it is called _____.
3. A fuse is always connected in _____ with the circuit to be protected.
4. A fuse element should have _____ melting point.
5. The fuse wire having low-melting point is an alloy of _____.
6. The minimum value of current at which the fuse element melts is called _____.
7. The basic purpose of earthing is to _____.
8. For a good earthing system, the value of earth resistance should be _____.

**OBJECTIVE TYPE QUESTIONS**

1. In motor wiring installations, double earthing is required.
(a) True (b) False
2. The material used for fuse element is
(a) copper. (b) aluminium. (c) tin-lead alloy. (d) Any of (a), (b), or (c)
3. The material used for fuse must have
(a) low-melting point and low-specific resistance.
(b) low-melting point and high-specific resistance.
(c) high-melting point and low-specific resistance.
(d) low-melting point with any specific resistance.
4. The rewirable fuses do not deteriorate with age.
(a) True (b) False
5. In fluorescent tubes, the outer glass tube is internally coated with
(a) quartz. (b) paint. (c) telecom powder. (d) phosphor.
6. Fuse is always connected
(a) in series with the circuit to be protected. (b) in parallel with the circuit to be protected.
(c) Either (a) or (b) (d) None of the above
7. Fuse is always connected in
(a) neutral. (b) earth. (c) phase. (d) Any of (a), (b), or (c)
8. The basic purpose of earthing is that
(a) it avoids faults.
(b) it allows the current to flow in the circuit.
(c) it protects the operator from electric shock.
(d) it stops current to flow in the circuit.
9. When more than one equipment is to be earthed
(a) parallel connections should invariably be used. (b) series connections should invariably be used.
(c) Either (a) or (b) (d) None of the above
10. For proper earthing, according to Indian electricity rules, of heavy power equipment,
(a) single earthing is sufficient, (b) double earthing system has to be adopted,
(c) half earthing is sufficient, (d) Any of (a), (b), or (c)
11. Earth resistance should be very low.
(a) True (b) False

**VIVA VOICE/REASONING QUESTIONS**

1. Electric shock is dangerous for a human body. Why?
2. All the metallic bodies or frames of all the electrical equipment/apparatus are properly earthed. Why?
3. Fuse is always provided in phase wire. Why?
4. MCB is always provided in phase wire. Why?
5. Double earthing is provided for all the electrical equipment operated at 400 V and above. Why?
6. Pipe earthing is preferred over other types of earthing. Why ?

**SHORT ANSWER TYPE QUESTIONS**

1. What do you mean by electric shock?
2. How one can prevent himself from electric shock?

3. If a person is electrocuted, what steps will you take?
4. What do you mean by earthing?
5. How earth resistance can be kept low in dry summer season?
6. What do you mean by double earthing?
7. Which type of earthing preferred in pipe earthing or plate earthing? Give reasons.
8. What is fuse wire?
9. How a fuse protects appliances?
10. A fuse wire should have low- or high-melting point. Explain the reason.
11. An MCB is just like a fuse or switch. Explain.
12. What is an ELCB? Mention its function.



TEST QUESTIONS

1. Give brief description of any five safety measures you will take while handling the electrical equipment.
2. What do you understand by electric shock? Explain how it damages the human body.
3. How a person is disengaged from electrical line in case of an accident?
4. Explain in brief the action you will take in restoring a person who has suffered an electrical shock and is unconscious.
5. What are the causes of electric fire?
6. How can we prevent electric fire?
7. How can we prevent electric shock?
8. What is meant by a fuse? Enlist the desirable properties of a fuse wire.
9. Write short note on a rewirable fuse.
10. How does a fuse protect and maintain the life of an electrical equipment?
11. Write short note on HRC (or HBC) fuse.
12. Enumerate the disadvantages of a rewirable fuse.
13. What is earthing?
14. What is the purpose of earthing?
15. Explain system earthing and equipment earthing.
16. What do you understand by double earthing?
17. What are the various methods of earthing? Explain pipe earthing.



ANSWERS

Fill in the Blanks

- | | | | |
|-----------------|-------------------|-------------------------|---------------------------------|
| 1. 10 mA | 2. fuse | 3. series | 4. low |
| 5. tin and lead | 6. fusing current | 7. protect the operator | 8. very small below $1\ \Omega$ |

Objective Type Questions

- | | | | | |
|---------|--------|--------|--------|---------|
| 1. (a) | 2. (d) | 3. (a) | 4. (b) | 5. (d) |
| 6. (a) | 7. (c) | 8. (c) | 9. (a) | 10. (b) |
| 11. (a) | | | | |